MyDAW Operation Manual

From your first recording to editing, mixing and exporting

Version 4.0

MyDAW is Mac software (a DAW) for recording sound from a microphone or audio interface and for playing back, editing and mixing many tracks together.
This manual follows the order in which you actually work, so that even first-time DAW users will not get lost. Start by reading chapters 2 to 5 in order while you record your first song.
The screens can be shown in English or Japanese (chapter 17). This manual describes the English screens.

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1. What you can do with MyDAW

- **Record** in mono or stereo on as many tracks as you like, at 24-bit and 44.1, 48, 88.2 or 96 kHz.
- **Overdub**: record a new track while listening to tracks you have already recorded.
- **Punch in**: re-record just the part that went wrong. The joins are smoothed automatically.
- **Edit** recordings (clips): move, trim, split, duplicate, change volume, fade (with adjustable curves), normalize and reverse. Select several clips or a time range at once and cut, copy and paste. Everything can be undone.
- **Organise** tracks in folders, which move, mute, solo and fold as one.
- **Tune** your guitar before recording with the built-in tuner.
- **Mix** with faders, pan, mute, solo and meters.
- **MIDI**: play and record from a MIDI keyboard or import MIDI files to sound software instruments (AU instruments); edit notes in the piano roll, quantize them and adjust velocities (chapter 12).
- **Use effects**: Audio Unit (AU) and VST3 plug-ins such as compressors, EQs, reverbs and amp simulators.
- **Export** the finished mix as one WAV or MP3 file.

2. Getting ready

What you need

- A Mac with Apple Silicon (macOS 13 or later)
- A microphone or an audio interface (the Mac's built-in microphone also works)
- Headphones (if you use speakers while recording, the microphone picks them up)

Start MyDAW and choose a project

1. Double-click `MyDAW.app`.
2. On the start screen choose **New Project** or **Open Project** (to continue earlier work).
3. For **New Project**, choose a folder in the save panel, type the project name in **Name** and click **Create**. MyDAW creates a `.mydaw` file with that name and a `Recordings` folder for your audio in the chosen folder.
4. For **Open Project**, choose the `.mydaw` file of an existing project.

A folder for each project Click **New Folder** at the bottom left of the save panel to make a folder for the project there and then. The new folder is selected, so just type the name and click **Create**. If you choose the Desktop or a similar place directly, the `Recordings` folder is created there.

Recent Projects The **Recent Projects** list below the buttons shows up to 50 projects you have opened or saved, newest first. Click a project name (in orange) to open it without choosing its file. The date on the right is when it was last saved.

First launch only When macOS asks whether “MyDAW” may access the microphone, choose **Allow**. Without it you cannot record.

Plug-in scan Right after launch, a log shows MyDAW looking for installed plug-ins. From the second launch on this takes almost no time.

One-time setup

Click the gear button ⚙ at the top to open **Settings** and select your equipment (details in chapter 17).

1. Choose your audio interface for **Input device** and **Output device**.
2. Set **Buffer size** to **256** to start with.
3. Click **Apply**. If you changed a device, choose **Save and Restart** when MyDAW asks to restart.

The macOS input and output change too While it runs, MyDAW also makes the devices chosen here the macOS input and output (System Settings → Sound), and puts the previous ones back when it quits. Do not change the input or output in System Settings while MyDAW is running.

Screen language At first the screens follow the macOS language (Japanese or English). You can choose “English” or “日本語” under **Language** in Settings (chapter 17).

3. The main window

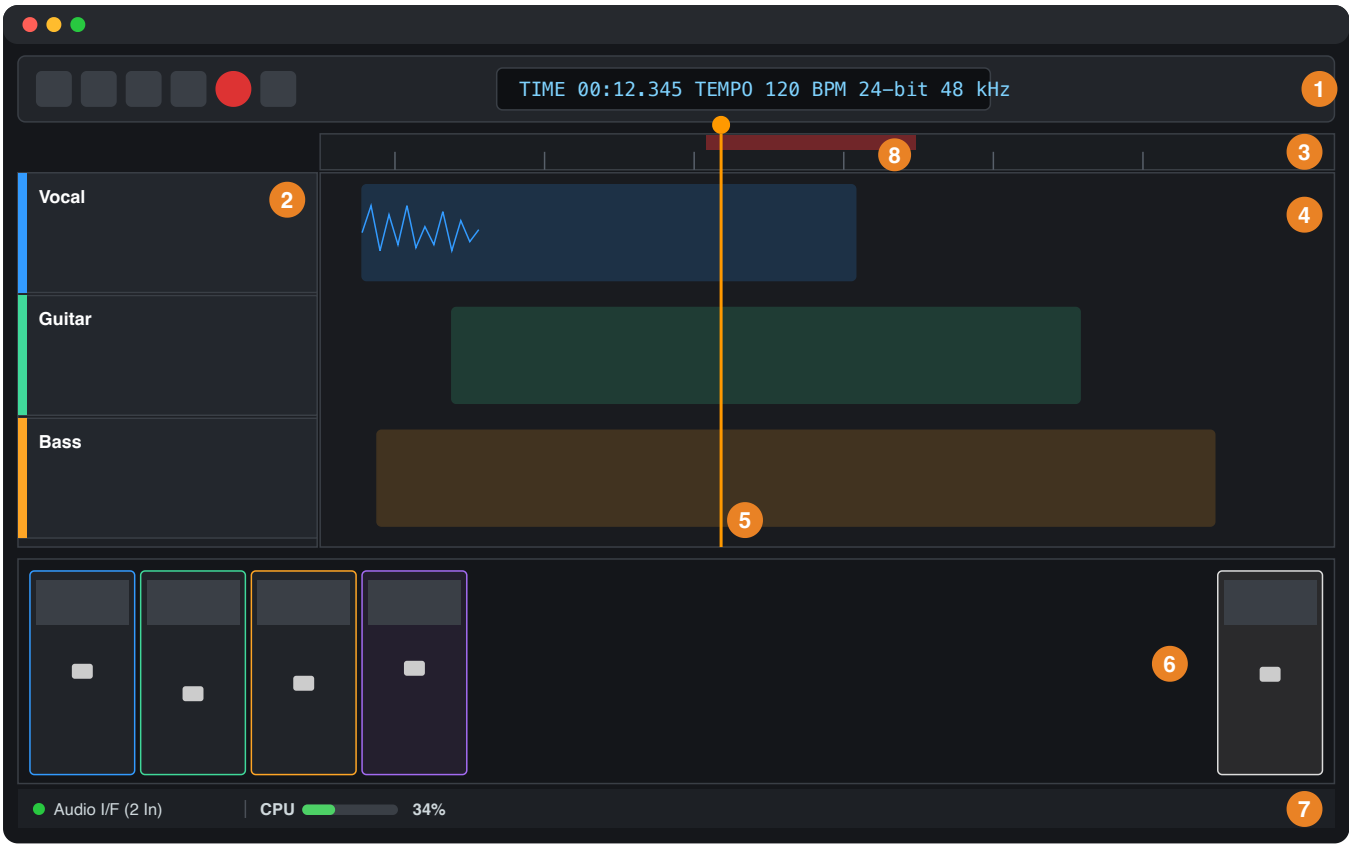


Figure 3-1 Layout of the main window

1	**Transport bar**: buttons for play, record, save and so on, plus time and tempo displays.
2	**Track headers**: track name, record arm, mute/solo, input selection and more.
3	**Ruler**: time (or bars and beats). Click it to move the playback position there.
4	**Track lanes**: where your recordings (clips) appear as waveforms.
5	**Playhead (line and ●)**: the current playback / recording position. Orange when stopped, green while playing, red while recording (green while waiting for a punch-in). While the transport runs, the ● bounces on every beat.
6	**Mixer**: volume, pan and effects for every track. Drag its top edge to change its height (at least 220 pt always stays between the send/fader divider and the bottom edge). The ▼ on its bar folds it away (chapter 9). The **Mixer** / **Piano Roll** tabs on its bar switch to the piano roll for editing MIDI (chapter 12).
7	**Status bar**: the audio device in use, MIDI devices and CPU (audio processing load).
8	**Punch range**: the part re-recorded with punch-in (chapter 7).

The open project's name (the file name without `.mydaw`) is shown in the middle of the bar at the very top of the window, where the close button is.

CPU meter and dropout mark

CPU in the status bar shows how much headroom audio processing has left: the time taken to produce one buffer of sound as a share of how long that sound lasts. Near 100% processing can no longer keep up and the sound breaks up (a dropout).

- **Bar colour**: green while there is room, yellow above 60%, orange above 80%, red at 100%.
- ● **Dropout**: a red mark shown for 3 seconds when processing fell behind and the sound broke up.
- **Details**: hover over the CPU meter to see the average and peak over the last second, MyDAW's CPU usage across all cores, and how many dropouts there have been since launch and when the last one was.

> **Why it differs from Activity Monitor** Activity Monitor shows how much work the whole Mac is doing. The CPU meter shows whether each buffer is finished in time, so it can be high while the Mac as a whole has room to spare. This is the value that matters for dropouts.

Transport bar



Figure 3-2 Transport bar (hover over a button to see its name)

No.	Button	What it does
1	Settings	Opens Settings (audio devices, language and more; chapter 17).
2	Undo	Undoes the last clip edit (`⌘Z`).
3	Redo	Redoes an undone edit (`⇧⌘Z`).
4	Rewind	Returns to the start, 00:00.000 (`←`). With a song start flag, it first returns to the flag (chapter 13).
5	Play / Pause	Starts and stops playback (`Space`). **This button does not record.**
6	Record (red circle)	Starts recording on record-armed (R) tracks (`R` key).
7	P	Turns punch in/out on (chapter 7). It cannot be pressed from the moment you click Record until you stop; with punch on, P blinks meanwhile before (and after) the punch range and stays lit while recording inside it.
8	Rollback (↺)	Turns rollback recording on: recording starts playing a few bars before the playhead (chapter 4). The number is the bars to roll back (set in Settings). Shown dimmed while punch is on, because punch recording does not use it.
9	Metronome	Click on/off (also while playing or recording). **Right-click** it for a volume fader that changes the click volume, also while playing.
10	Tuner (tuning fork)	Opens the tuner. Press it again, or the window's close button, to close it (end of chapter 11).
11	Save	Saves the project (`⌘S`).
12	Open	Opens a project (`⌘0`).
13	Snap	Makes clip moves and trims snap to beats.
14	Auto-scroll	Turns on/off the view following the playhead while playing or recording (orange when on). When off, the view stays put even if the playhead leaves the screen.
15	Displays	TIME (position), TEMPO (type the BPM), METER (time signature, chapter 13), FORMAT (recording format and sample rate).
16	View scaling	Horizontal zoom, track height, waveform height (chapter 14).
17	Ruler mode	Switches the ruler between time and bars/beats.

Track header

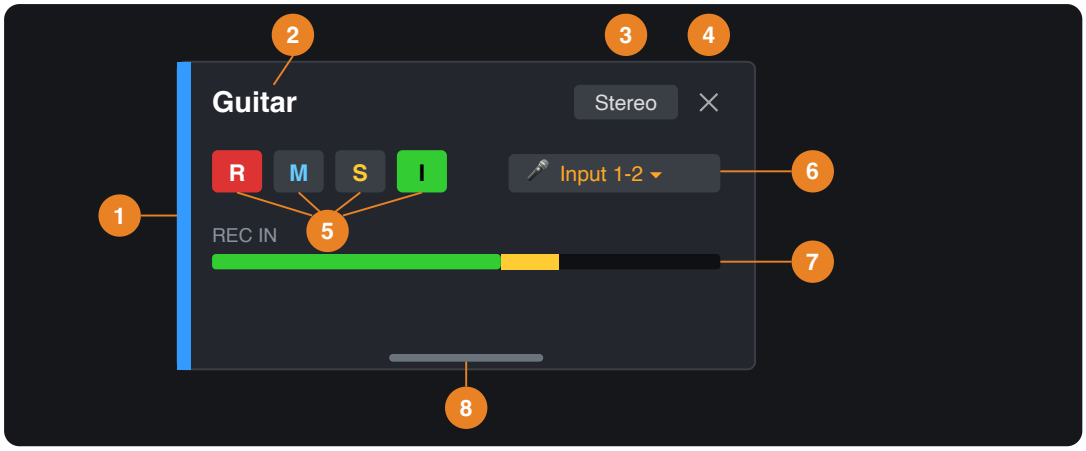


Figure 3-3 Track header

1	**Colour bar**: click to open a colour palette and change the track colour; “Custom…” lets you pick any colour.
2	**Track name**: double-click to rename. The current track's name is shown black on white.
3	**Mono / Stereo**: the track's channel count. Can be switched even on a track that already has recordings. While recording, it is locked (greyed out) on record-armed (R) tracks only. The recording files are not changed: on a mono track, stereo clips play with L and R summed at half level each; on a stereo track, mono clips play on both L and R. Switching back restores the original sound.
4	✕: deletes the track after you click **Delete** in the confirmation message. Deleting a track cannot be undone with `⌘Z`.
5	**R** record arm, **M** mute, **S** solo (hear only this track), **I** input monitoring (chapter 8). M or S lit grey means the track's folder has it on; it cannot be pressed then. R cannot be pressed while recording, and a track being recorded cannot change its input or be deleted. An armed track has a light red background in its header and waveform area, and its R blinks red while it waits to record; while it records, R is lit red.
6	**Input channel**: the audio interface input used for recording.
7	**Meter**: input level while armed, output level otherwise.
8	**Height**: drag the bottom edge to change the track height. Making it lower hides the meter first; it can shrink until only the track name and buttons show.

To add a track, click the **+** button above the track headers and choose **Add Audio Track**. The track is added below the selected track (the current track). For **Add Folder** in the same menu, see "Grouping tracks in folders" below.

Changing the track order

Grab a free spot on a track header (around the track name or the meter) and drag it up or down. While you drag, the track follows the mouse and a white line shows where it will land. Release the mouse to drop it there.

The mixer channels follow the same order (they cannot be reordered in the mixer itself). A drag cannot start on a button, a menu or the bottom edge (height). Reordering tracks cannot be undone with ⌘Z.

Grouping tracks in folders

With many tracks, grouping them by part (guitars, keyboards, …) in folders keeps the list easy to read.

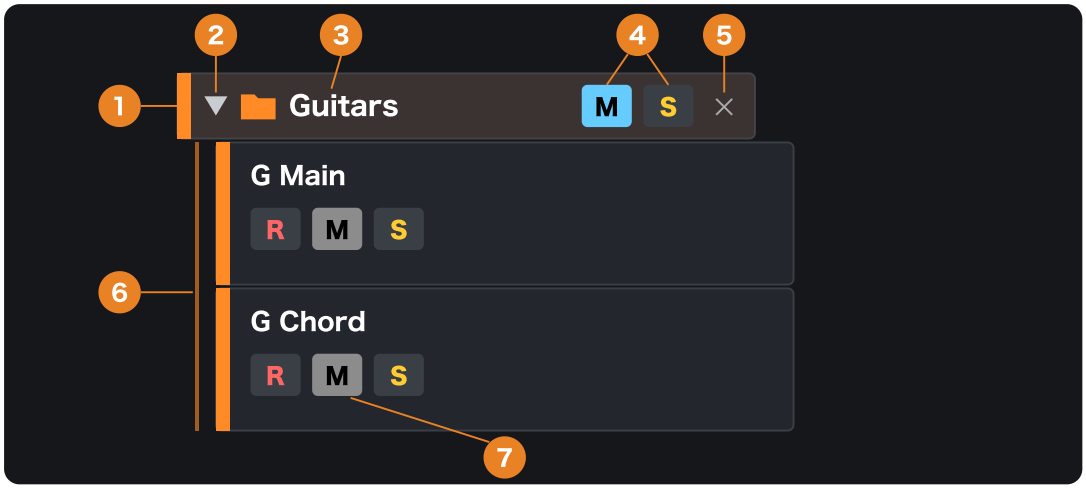


Figure 3-4 A folder and the tracks in it

1	**Colour bar**: click to change the folder's colour.
2	▼ / ▶: closes (▶) or opens (▼) the folder. The tracks of a closed folder are not shown, but they still play and record.
3	**Folder name**: double-click to rename.
4	**M / S**: mutes / solos every track in the folder, whatever the tracks' own buttons say. Press again and each track goes back to its own state. Not available while the folder is empty.
5	✕: deletes the folder after you confirm with **Delete**. Its tracks are not deleted; they stay where they are, out of the folder.
6	**Tracks in the folder**: shown indented, with a line in the folder's colour on their left.
7	**Grey M / S**: on because of the folder's M / S; they cannot be pressed until the folder's button is turned off.

- **Making a folder**: **+** → **Add Folder**. The folder is placed above the current track (above its folder when the current track is in one). Folders cannot be put inside folders.
- **Moving tracks in and out**: drag the track. When the white drop line is indented, the track goes into the folder. Below a folder's last track, dropping on the lower half of that track puts it inside, and on the upper half of the row below puts it outside. Dropping on a closed folder puts the track at the end of that folder.
- **Moving a folder**: drag the folder's row; its tracks move with it.
- Clicking a folder does not make it the current track. A track added inside a folder takes the folder's colour.

Header right-click menu

Right-clicking a track or folder row shows the menu below (duplicating is not available while playing or recording). Right-clicking a track makes it the current track.

Item	What it does
Add Audio Track	Adds a track above the clicked track (inside the same folder for a track in a folder). For a folder, adds the track as the folder's first.
Add Folder	Adds a folder above the clicked track or folder. Not shown for a track inside a folder.
Duplicate Track	Shown for a track. Adds a copy right below it with the same settings: clips, mixer state (volume, pan, M/S, sends) and plug-ins with their settings. A number is added to the name (“Guitar” → “Guitar 2”, “Guitar 2” → “Guitar 3”; the next free number if that name exists). The copy becomes the current track. Recording files are not copied; the copy uses the same files.
Duplicate Folder	Shown for a folder. Copies the folder and its tracks and adds them below the folder's last track. Names follow the same rule as Duplicate Track.
Show in Mixer	Scrolls the mixer so the track (for a folder, where the folder starts) is at its left edge.

4. Your first recording

Here you record one track and listen back to it.

1. **Choose a track**: click its `R` button so it turns red (armed). The track's background turns light red.
2. **Choose the input**: in the input channel menu (figure 3-3, item 6) pick the input your microphone or guitar is connected to. For a single voice or guitar, set the track to `Mono`.
3. **Check the level**: play and watch the meter. The loudest parts should reach about the yellow area. If it reaches red, turn down the input gain on your audio interface.
4. **Choose where to start**: click the ruler, or press `←` to start from the beginning.
5. **Record**: click the red **Record** button (or press `R`) and perform. The waveform is drawn as you go.
6. **Stop**: press `Space`. The recording becomes a clip.
7. **Listen back**: turn off `R`, press `←`, then `Space`.

Recording along with a backing After the first track, add a new track, arm it and record in the same way. You will hear the first track while recording the second. While recording, older clips on the track being recorded are silenced.

Starting with a run-up (rollback recording) Turn on the rollback button (figure 3-2, item 8) and recording starts playing the set number of bars before the playhead, so you can play along into the take. The recording itself starts at the playhead, and the clip begins there. The run-up is recorded too: drag the clip's left edge to the left to bring it back. Set the number of bars in Settings (**Rollback when recording**). When punch in/out is on, the punch range is used instead.

Recording in time Type the BPM into TEMPO and turn on the metronome before recording to play along with a click (chapter 13).

Recording files Recordings are saved as WAV files in the `Recordings` folder. The file is named after the track and a take number, such as `Bass_001.wav`.

5. Playback and basic controls

To…	Do this
Play / stop	`Space` or the Play button
Go back to the start	`←` or the Rewind button
Jump to a position	Click the ruler (also while playing)
Hear only some tracks	`S` (solo) on those tracks
Silence some tracks	`M` (mute) on those tracks

When the playhead reaches the right edge during playback, the view scrolls on automatically (turn this on or off with the Auto-scroll button in the transport bar). When stopped, drag the knob (●) of the scroll bar below the lanes to move around freely (clicking elsewhere on the bar does nothing).

6. Editing clips

One recording is called a **clip**. Editing never changes the original recording file (non-destructive editing), so feel free to experiment.

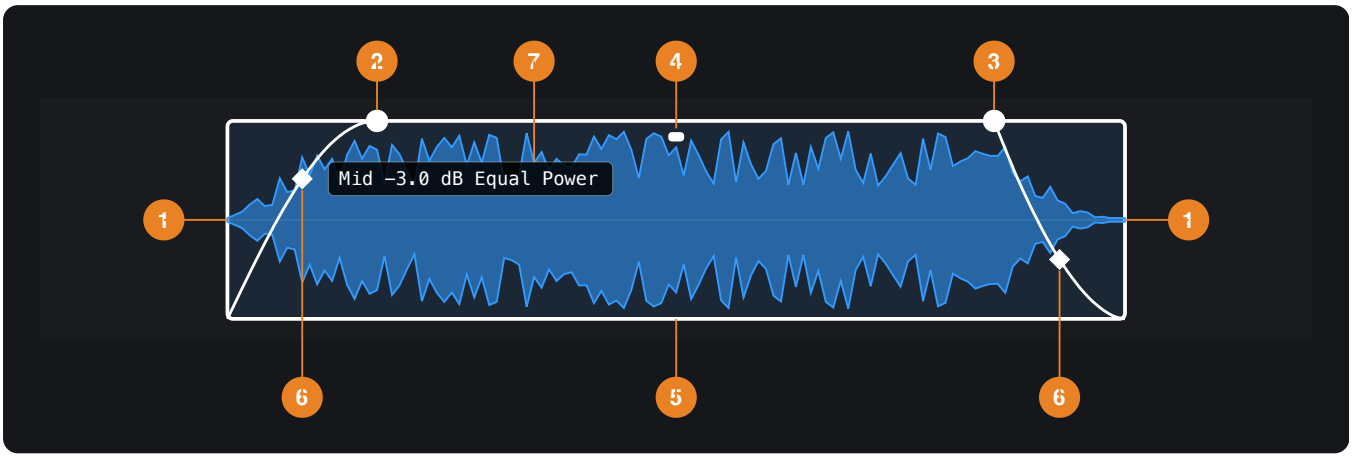


Figure 6-1 Clip handles

1	**Left / right edges (trim)**: move the pointer onto the clip's left or right edge line and it turns into an arrow (pointing right → at the left edge, left ← at the right edge); drag to shorten or lengthen the audible part.
2	**Fade in**: drag the white dot at the top left to the right so the sound starts gradually. Over the dot, the pointer turns into a pointing hand.
3	**Fade out**: drag the white dot at the top right to the left so the sound ends gradually (the pointer turns into a pointing hand, as for the fade in).
4	**Gain**: drag the short bar at the top centre up or down to change the clip volume (−∞ to +36 dB; below −72 dB it becomes −∞, silence). Over the bar, the pointer turns into an up-down arrow.
5	**Body**: click to select (thick white frame), drag to move — also onto another track.
6	**Fade curve**: the white diamond in the middle of each fade line (the pointer turns into a pointing hand over it). Drag it up or down to change the fade's shape (see "Shaping a fade" below).
7	**Value readout**: while you hold handle 2, 3, 4 or 6, a small box shows the fade length, the clip volume (dB) or the curve shape.

Waveforms are drawn at the level you will actually hear: they shrink along the fade curves, and raising or lowering the gain makes them taller or shorter.

Shaping a fade

Drag the diamond in the middle of a fade line (figure 6-1, item 6) up or down to bend the fade. The readout shows the level at the exact middle of the fade.

Diamond position	Fade shape
Moved up	Bowed up: a fade-in rises quickly; a fade-out holds its level and drops late.
Middle at -3.0 dB (**Equal Power**)	Keeps the perceived level steady — the best shape for crossfades. The diamond snaps here when close.
Middle at -6.0 dB (**Linear**)	A straight line. The diamond snaps here when close.
Moved down	Sagging: a fade-in rises slowly; a fade-out drops early.

Double-click the diamond to return to **Auto**: linear for a fade against silence, equal power for a crossfade over another clip. If a fade is too short to show its diamond, zoom in horizontally.

Selecting clips

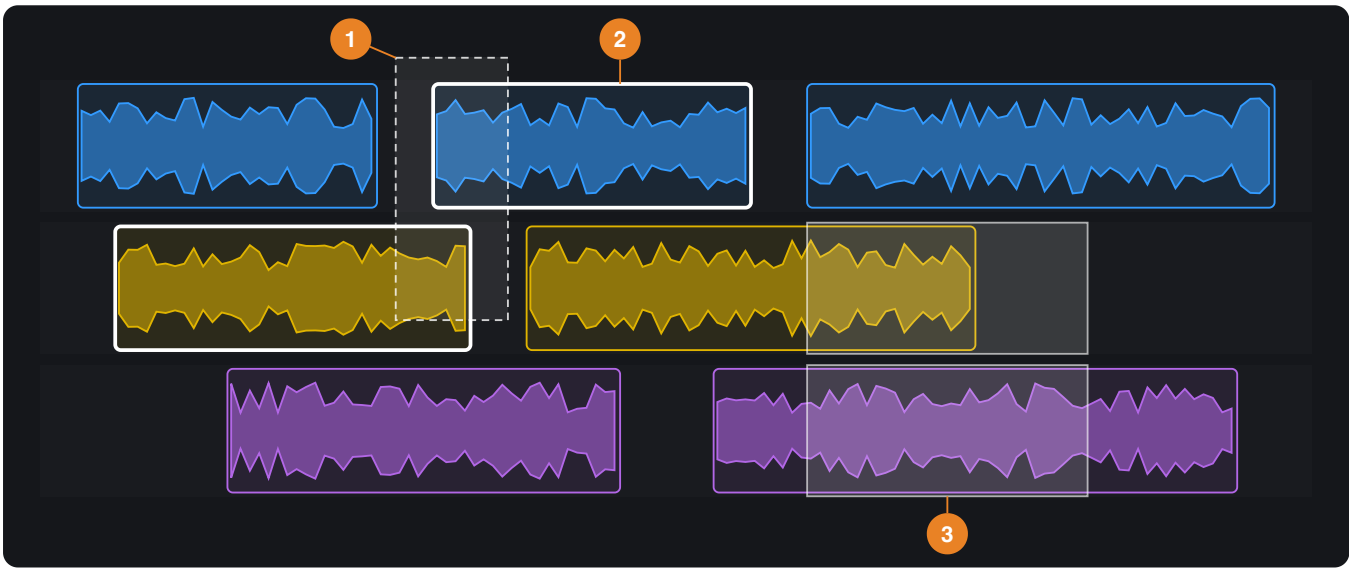


Figure 6-2 Marquee (1), the clips it selected (2), and a range selection (3)

To…	Do this
Select one clip	Click it
Add to / remove from the selection	⇧-click or ⌘-click
Select several at once (marquee)	Drag from an empty spot to draw a frame (figure 6-2, item 1). Every clip the frame touches is selected (item 2), across tracks too.
Add with a marquee	⇧-drag
Select everything	⌘A
Clear the selection	Click an empty spot, or press `Esc`

Dragging any selected clip moves all selected clips together as one block (also up or down to other tracks). If you release where any of them would have no track to land on (for example below the last track), they stay on their tracks and move in time only.

Selecting a time range

Instead of whole clips, you can select “from this second to that second”. Use it to remove or keep just part of a clip.

1. Hold ⌘ and drag from where the range should start to where it should end. You can start on a clip or on an empty spot.
2. The range is shown as a white band (figure 6-2, item 3). Drag up or down to extend it over several tracks.
3. Then use one of these:

Action	Result
`Delete` (or right-click → **Delete Range**)	Removes the audio inside the range (leaving silence). Clips are split at the range edges.
Right-click → **Crop to Range**	Keeps only the audio inside the range and removes what is before and after it.
Right-click → **Split at Range Edges**	Splits clips at both ends of the range.
⌘C / ⌘X	Copies / cuts only the audio inside the range.

Cut, copy, paste and duplicate

- ⌘X (cut), ⌘C (copy): take the selected clips (or, with a range selected, the audio in the range).
- ⌘V (paste): pastes at the **playhead**. Material copied from several tracks keeps its layout, with its top track landing on the **selected track**.
- ⌥-drag a clip: leaves a copy in place while you move it (applies to every selected clip).
- Cut / Copy / Paste at Playhead are also in the right-click menu.

> **While typing a name** When you are typing in a text field (such as a track name), ⌘X, ⌘C, ⌘V and ⌘A edit the text instead.

Right-click menu

Right-clicking a clip that is not selected selects it. Right-clicking a selected clip keeps the selection, and the menu commands apply to **all selected clips** (with several, the items show the count, such as “(3)”). Right-clicking inside a selected range keeps the range and shows the range menu.

- **File name**: the top of the menu shows the name of the clip's audio file, followed by its channels, bit depth, sample rate and size (for example `Bass_001.wav, mono 24bit 44.1kHz 7.8MB`). Choose it to show the file in the Finder (with several clips selected it shows the count, such as “3 Clips”). New recordings are named “*track name_number*.wav”, the number going up by one with each take.
- **Cut / Copy / Paste at Playhead**: cut, copy, and paste at the playhead.
- **Delete Range / Crop to Range / Split at Range Edges**: shown only while a range is selected.

- **Normalize**: sets the clip's gain so its loudest sample reaches 0 dBFS (the 24-bit maximum). The whole recording is measured, including parts hidden by trimming. The audio file is not changed (the gain limit is +36 dB).
- **Reverse**: plays the clip backwards. A reversed WAV file (such as `Reverse_Bass_001.wav`) is written to `Recordings` and the clip switches to it. Fade lengths and shapes swap ends as well.
- **Strip Silence…**: removes the parts where the audio stays at or below the silence level for at least the given number of seconds. Useful for clearing the unrecorded parts of stems exported from another DAW. In the dialog, set the silence level (dB, −72 dB at first), the minimum silence length (seconds) and the fade in/out (ms, 10 ms at first). Any single sample above the silence level counts as sound, so for stems with dither or faint noise, set the level above it. To remove only digital zero, use −144 dB. The clips are split around the silent parts, the silent pieces are deleted, and the new clip edges get fades. The fades sit outside the sound (in the silence), so the starts and ends of the sound are not cut into. A fade longer than half the minimum silence length is shortened to half of it. The audio files are not changed.
- **Choose Recording File…**: picks the file again if it cannot be found. Available when a single clip is selected.
- **Mute / Unmute Recording**: silences the clips. If any of them is not muted the item mutes them all; when all are muted it unmutes them.
- **Duplicate Recording**: copies the selected clips as one block whose start lands on the playhead (spacing and tracks kept). The copies become the selection.
- **Split at Cursor**: splits the clips the playhead is inside in two. Unavailable when the playhead is inside none of them.
- **Delete Recording**: deletes it.

Snap

With Snap on (figure 3-2, item 13), clip moves, trims, range selections, moving the playhead by clicking the ruler, and moving the song start / end flags snap to whole beats. Turn it off for fine positioning.

Undo and redo

Moves, trims, fade lengths and shapes, gain, delete, split, duplicate, range edits, cut and paste, normalize, reverse, recordings and WAV imports can be undone with `⌘Z` and redone with `⇧⌘Z` (while stopped). Undoing a recording or an import removes the clip; the file stays in the Recordings folder.

7. Punch-in recording and layered takes

Use this when a take is almost perfect except for a few bars: re-record only that part.

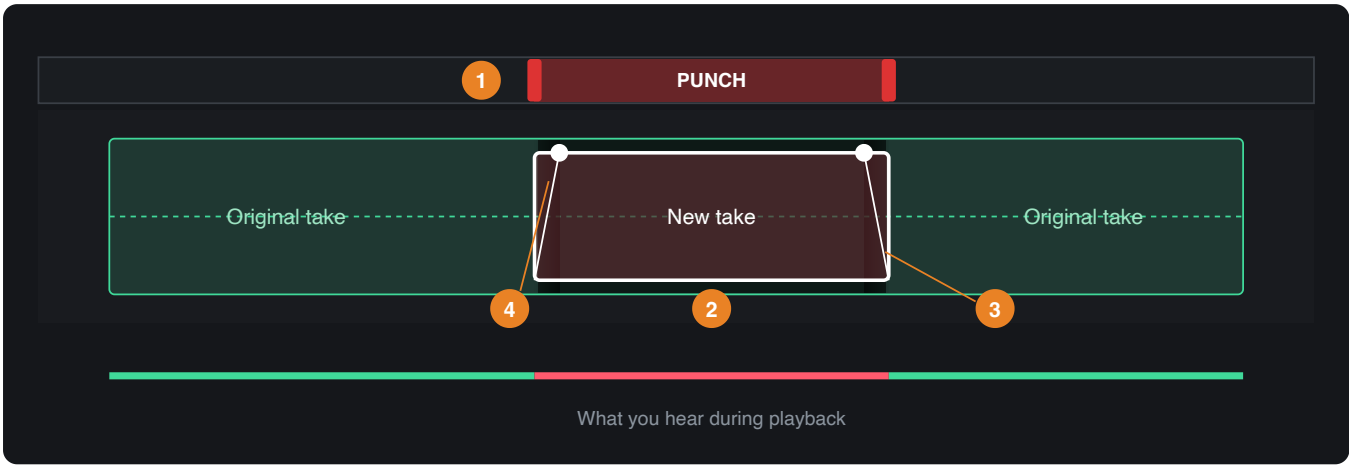


Figure 7-1 Punch range and the re-recorded part on top

1. Drag the red handles of the **punch range** on the ruler (figure 7-1, item 1) to the part you want to redo (they move in beats).
2. Click `P` in the transport bar to enable punch in/out (the range turns red).
3. Turn on `R` on the track to redo.
4. Click Record a little before the range and perform. Only what falls inside the range is used, and only that part is shown while you record. The punch range on the ruler blinks before (and after) the range and stays lit while recording inside it. `P` does the same: it blinks outside the range and stays lit while recording inside it. P cannot be pressed until you stop.
5. When you stop, the new part appears as a new clip (item 2) on top of the original clip.

How overlapping clips play

- Where clips overlap, **only the clip recorded later (the upper clip) plays**. The hidden part of the original clip is shown darker, with a flat (silent) waveform.
- The joins automatically get a **crossfade** (item 3), so there are no gaps or clicks. The default is 10 ms with an equal-power shape. Across the crossfade the upper clip's waveform grows while the lower clip's shrinks.
- Moving the fade handles of the upper clip (item 4) changes the crossfade length, and the diamond in the middle of the fade line changes its shape (chapter 6). The lower clip automatically gets the same shape mirrored. The lower clip's fade on the overlapped side cannot be edited.
- If you move or delete the upper clip, the lower clip automatically plays normally again.
- You can punch in as many times as you like over the same place. The last take is always on top, and every join gets its own crossfade.

> **Fine-tuning the range** The whole pass is saved, including audio outside the range. Drag the edges of the upper clip to extend the re-recorded part.

> **This rule applies to every overlap** Not only punch takes: whenever clips overlap after moving or duplicating, the clip on the higher layer wins.

Clip layers (which clip is on top)

- The layer order is **the order in which clips were added to the track**: the later a clip was recorded, the higher it is.
- Dragging a clip within the same track does **not** change its layer. For example, you cannot bring the first punch take above the second by moving it.
- Clips made with **Duplicate Recording** or paste (`⌘V`), and clips moved to another track, are placed on **top** of that track.
- With `⌥`-drag, the copy left behind sits directly below the clip you moved.

8. Recording through effects (input monitoring)

Use this, for example, to hear your guitar through an amp simulator while recording.

1. Insert the effect you want (e.g. an amp simulator) on the track (chapter 10).
2. Turn on `R` and `I` on the track. When I is filled green you hear the input through the effects (also while stopped).
3. Record with the Record button as usual.

Turning I on while playing Input monitoring cannot be rewired during playback. If you turn I on while playing, you start hearing the input after you stop, once reverb and other tails have faded (usually 1–2 s, at most 8 s). If you turn I off while playing, the input stays audible until you stop. Switching I while stopped takes effect at once.

The dry signal is recorded The file contains the sound before the effects, so you can change the amp settings later.

If you hear a delay Set **Buffer size** to 128 or 256. If your audio interface's direct monitoring is on, you will hear the dry and processed sound doubled — turn direct monitoring down for that input.

Feedback With a microphone and speakers you may get feedback (a loud squeal). Use headphones.

9. Using the mixer

The mixer at the bottom sets each track's level, left–right position (pan) and effects. Tracks come first (in the same order as the track headers), then FX channels, with MASTER (the overall output) at the right. When the channels do not fit, drag the knob (●) of the scroll bar below them to move sideways (as in the track view, clicking elsewhere on the bar does nothing).

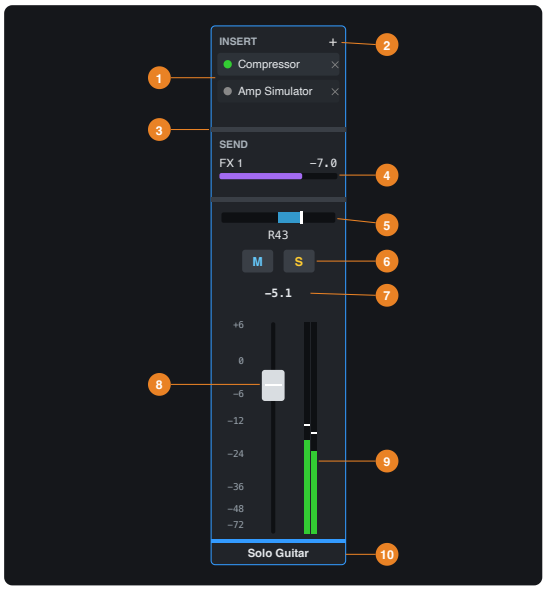


Figure 9-1 One mixer channel (strip)

1	**INSERT (plug-in list)**: green dot turns a plug-in on/off, click the name to open its window, drag to reorder (drag to another channel to copy), ✕ removes it.
2	**+**: adds a plug-in.
3	**Section divider**: drag up or down to change the height of the INSERT, SEND and control sections (all channels together).
4	**SEND**: how much goes to an FX channel. Drag sideways, or double-click the value to type it.
5	**Pan**: drag sideways to place the sound left or right. `<C>` is centre, `L56` left, `R75` right.
6	**M / S**: mute and solo. FX channels have them too.
7	**Level value**: current fader value in dB. Double-click to type (e.g. `-6`, `-inf`).
8	**Fader**: drag up or down to change the level. 0 dB is the reference; you can go up to +6 dB.
9	**Meters (L / R)**: level of each side. Green → yellow (from -12 dB) → red (from -6 dB). The white line shows the recent peak.
10	**Name**: click to select the track (clicking anywhere on the channel outside its controls does too). Double-click an FX channel's name to rename it. The current track's name is shown black on white.

Operating several channels together

Other tracks' channels can be operated together with the current track.

1. Click the channel to work from in the mixer, making it the current track.
2. ⇧- or ⌘-click the channels to operate with it. Add as many as you like; click one again to take it out.

- The channels operated together get a lighter background. Only the current track's name is shown black on white.
- Moving the fader or a send of any of them moves the others by the same number of dB; pan moves them by the same amount. A fader or send at -∞ follows only when the one you move also starts at -∞.
- Pressing **M** / **S** on any of them sets the others to the same state.
- Inserts (plug-ins) change only on the channel you click.
- A plain click on a channel, or making another track current (in the track headers, for example), ends the grouping. The track headers and waveforms do not show it.

A vertical line in the folder's colour stands left of each folder's tracks, and one in the FX colour left of the first FX channel. Click a line to open the colour palette: it changes the folder's colour (the folder in the track headers follows), or the colour of every FX channel.

Handy actions

- ⌥-click: resets faders and sends to 0 dB and pan to centre.
- ⌘-drag: fine adjustment.
- Drag the mixer's top edge: changes the height of the whole mixer.
- ▼ on the mixer's bar: folds the mixer down to its bar; ▲ brings it back.
- Right-click a track header → **Show in Mixer**: scrolls that track's channel to the left edge (unfolding the mixer if needed).
- Right-click in the mixer: **Add FX** adds an FX channel.
- Right-click an FX channel strip: **Remove (channel name)** removes that FX channel.

Output destinations and buses

The pop-up under a channel's fader chooses where its sound goes. It starts at MASTER.

To control several channels with one fader, use a bus.

1. Click **Add Bus** at the top of the mixer (or right-click in the mixer). A bus channel appears.
2. In the pop-up of each channel you want to group, choose that bus.
3. Set the overall level with the bus fader. Buses can have plug-ins too.

- A bus has an output destination too. Choices that would send the sound back to where it came from are not offered.
- To remove a bus, right-click its channel and choose **Delete Bus**. Channels going to it then go to MASTER.
- Add and remove buses and change destinations while stopped.

Volume automation

The volume can change by itself during the song. It works as a separate level control: the fader stays where it is. Tracks and MASTER both have it.

1. Press **A** in the track header. For MASTER, press **A** on the MASTER channel in the mixer. On a track, a blue line appears over the waveforms; for MASTER, a MASTER row appears below the track list with the blue line in it.
2. Click the line to add a blue dot at that place.
3. Drag a dot to set the level (up and down) and the position (left and right). The dB value is shown while you drag. With snap on, the position follows the grid.

- Right-click a dot for **Enter Value…** (+12 dB down to -∞) and **Delete**.
- The display next to **A** shows the level at the playhead.
- Turning **A** off hides the line and stops the level changes. The dots are kept, so turning it back on restores them.
- Dot edits can be undone with `⌘Z`. Dots do not move with clips.
- MASTER automation comes after the Post effects, so exports include it. It also affects the metronome and input monitoring.

> **Level tip** Adjust the track faders so the MASTER meter does not reach red. If the mix is too loud, lower the tracks rather than MASTER.

10. Effects (plug-ins) and FX channels

Inserting an effect

1. Stop playback (plug-ins cannot be inserted while playing).
2. Click `+` in the channel's INSERT section and choose a plug-in.
3. Click the plug-in's name to open its window.

Plug-in windows stay in front of MyDAW's window. When you switch to another app they stay on screen, behind that app and in front of MyDAW's window (the tuner window too).

Sound passes through the plug-ins from top to bottom. Drag the names to change the order. Plug-ins on MASTER's **POST** section affect the whole song (a limiter, for example).

Dragging a plug-in name to the INSERT list of **another channel** inserts a copy with the same settings (the original stays where it is). Drop it on a plug-in name to place it before that plug-in, or on an empty part of the list to add it at the bottom. This is handy for using the same effect settings on several tracks.

The green dot left of a plug-in name turns the plug-in off (also during playback). A plug-in that is off passes its input through unchanged. It keeps running while off, so turning it off does not lower the CPU load; remove it with ✕ to do that.

The effects that come with MyDAW ("MyDAW: …" in the list) are described in chapter 11.

> **Supported formats** Audio Unit (AU) and VST3 effects. If a product is installed in both formats, only the AU version is listed (using both at once is unstable). Instruments are not supported.

> **When a plug-in name is shown in red** That plug-in cannot be used. Hold the pointer over the name to see why. "This plug-in cannot be used" means the plug-in is on this Mac but cannot run in MyDAW (such as a mono-only "(m)" version; use the stereo "(s)" version). "This plug-in was not found" means it is not installed on this Mac (for example, a project made on another Mac). A red plug-in does not affect the sound, and you can remove it with ✕.

FX channels (sharing a reverb)

Spatial effects such as reverb are usually placed once on an FX channel; each track then decides how much it sends (SEND).

1. Click **Add FX** at the top of the mixer (or right-click → Add FX).
2. Insert a reverb on the FX channel and set its Mix (dry/wet) to 100% wet.
3. Raise the SEND to that FX on each track.

A send carries the track's sound after its effects, fader and pan, so lowering a track's fader also lowers its reverb.

A new FX channel is named "FX 1", "FX 2", … with the number after the highest one in use, and takes the colour of the other FX channels (change it by clicking the line left of the FX channels; chapter 9).

Muting and soloing an FX channel

- **M (mute)**: silences the FX channel's output. A reverb tail stops at once.
- **S (solo)**: as with tracks, only soloed channels are heard.
  - Soloing a track also keeps the FX channels it sends to, so you hear its reverb.
  - Soloing an FX channel plays only that FX channel (the effect sound). The dry sound of the tracks sending to it is not heard, which is handy for checking the reverb on its own.

Removing an FX channel

1. In the mixer, right-click the strip of the FX channel you want to remove.
2. Choose **Remove (channel name)**, for example "Remove Reverb".
3. When asked “Delete “FX 1”?” (with the channel’s name), click **Delete** (or **Cancel** to keep it).

The plug-ins on that FX channel and every track's send to it are removed as well.

> **This cannot be undone** `⌘Z` does not bring back a removed FX channel. Save the project first if you may need it again.

Automatic delay compensation

Some plug-ins need time to process audio (latency). MyDAW reads the latency they report and adjusts playback timing automatically.

- Latency is compensated for plug-ins on tracks and on FX channels alike, so a reverb on an FX channel stays in time with the dry sound.
- Turning a plug-in off does not change the compensation (while off, it passes its input delayed by its own latency), so switching it during playback never interrupts the sound.
- If a plug-in's latency changes with its settings, the compensation follows.
- The sound right at the start position is heard too. With high-latency plug-ins, playback takes that much longer to get going after you press Play (for example 40 ms).
- A high-latency plug-in on an FX channel also delays input monitoring on armed tracks by that amount (the recording itself is not affected).

11. MyDAW's built-in effects

MyDAW comes with six effects ready to use. They appear in the insert `+` list as "**MyDAW: …**" (see chapter 10 for inserting effects). There is also a **tuner**, opened from the transport bar (end of this chapter).

Effect	What it does	Where to insert it
MyReverb	Plate reverb: adds space and ambience	On an FX channel, fed from tracks by sends
MyDelay	Delay: adds echoes	On an FX channel, fed by sends
MyChorusPan	Chorus, flanger and auto pan: movement and width	A track insert (on an FX channel with MIX at 100 %)
MyChannelStrip	4-band EQ and compressor to shape each sound	First insert of each track
MyDeEsser	De-esser: softens harsh consonants ("s" sounds) in vocals	On the vocal track, after MyChannelStrip
MyMaximizer	Maximizer: makes the whole mix louder without clipping	Last post insert on MASTER

Common layout and controls

Every effect shows, from the top:

- The **effect name**, with a mode display or switch on the right.
- The **channel name**: the track, FX channel or MASTER the effect is on. It follows when you rename the track.
- **IN / OUT meters**: the level going into and out of the effect. The number on the right is the highest peak so far; click it to clear it.
- **Faders and knobs**, each with its name and current value above it.

Action	Result
Drag a fader or knob	Changes the value (a click alone does not make it jump). Drag knobs up and down.
Drag with `⌘`	Fine adjustment.
`⌥`-click	Back to the default value.
Double-click the value	Type a value (`return` to apply, `esc` to cancel).

Every control can be recorded as automation, and the settings are saved with the project.

MyReverb (reverb)



Figure 11-1 MyReverb

A bright, dense reverb modelled on a metal plate. It spreads wide across the stereo field and starts from the side the track is panned to. It gives vocals and snares a natural sense of space.

Fader	Range (default)	What it does
HPF	Thru to 1 kHz (80 Hz)	Cuts lows from the reverb. Raise it to keep the low end clean and the mix clear. The bottom is Thru (no cut).
LPF	200 Hz to Thru (8 kHz)	Cuts highs from the reverb. Lower it for a darker, softer sound. The top is Thru (no cut).
RT	0.1 to 60 s (2.00 s)	Reverb time: how long the tail takes to fall by 60 dB. About 0.5 to 1 s for a small room, 2 to 3 s for a hall.
PD	0 to 1 s (20 ms)	Pre-delay: the gap between the sound and the start of the reverb. 20 to 40 ms keeps the sound from being buried in the reverb.
WIDTH	0 to 100 % (100 %)	Stereo width of the reverb. Lower it to pull the reverb towards the centre; 0 % is mono.
MIX	0 to 100 % (100 %)	Amount of reverb. Leave it at 100 % on an FX channel.

1. Add an FX channel and insert MyReverb on it (MIX at 100 %).
2. Raise the send of each track you want to add reverb to. The send sets how much reverb that track gets.

3. Set the length with RT and the start with PD. If the low end gets muddy, raise HPF.

> **Inserting it on a track** Lower MIX to about 15 to 30 %. At 100 % you hear only the reverb, not the original sound.

MyDelay (delay)



Figure 11-2 MyDelay

Repeats the sound after a delay (echo). Choose one of four modes from the menu at the top right; the faders shown depend on the mode.

Mode	Sound	Faders
Mono Delay	The same echoes on both sides	TIME, FEEDBACK, MIX
Stereo Delay	Left and right each get their own echoes (default)	TIME, FEEDBACK, WIDTH, MIX
Doubler	Delays the left by TIME and the right by 1.5 × TIME, once, to thicken and widen the sound	TIME, WIDTH, MIX
Ping-Pong	Echoes bounce between left and right	TIME, FEEDBACK, WIDTH, MIX

Fader	Range (default)	What it does
TIME	1 ms to 10 s (250 ms)	Time between echoes.
FEEDBACK	0 to 100 % (30 %)	How long the echoes go on. Even at 100 % they never keep getting louder.
WIDTH	0 to 100 % (100 %)	Stereo spread of the echoes; 0 % puts them in the centre.
MIX	0 to 100 % (100 %)	Amount of echo. Leave it at 100 % on an FX channel.

Following the tempo (SYNC): turn on **SYNC** above the faders and the time between echoes becomes a note length at the song's tempo (BPM). When you change the tempo, the echoes follow.

Control	What it does
SYNC	Lights orange when on; TIME then follows the tempo (off by default).
TIME (SYNC on)	Chooses the note length, from the bottom: 1/64, 1/32, 1/16, 1/8, 1/4 (default), 1/2, 1 bar, 2 bar, 4 bar (1 bar is one bar of the song's time signature).
N, **T**, **D**	Note type: N normal, T triplet, D dotted. TIME reads “1/8”, “1/8T”, “1/8D” and so on.
Readout at the right	The current tempo and the resulting echo time in ms. Example: 1/8D at 120 BPM is 375 ms.

Even with SYNC on, echoes are at most 10 seconds apart. In Doubler the right side stays at 1.5 × TIME.

Changing TIME or the mode during playback causes no clicks or pitch wobble.

MyChannelStrip (EQ and compressor)



Figure 11-3 MyChannelStrip

A 4-band EQ to shape the tone and a compressor to even out the level, in one effect. Insert it first on each track. When inserted, the EQ is flat (all bands at 0 dB) and the compressor is off, so the sound does not change. Parts that leave the sound unchanged are not computed, so it costs little even on many tracks.

Part	What it does
EQ ▸ COMP (top right)	Processing order. Click to switch to **COMP ▸ EQ**; the EQ and compressor sections swap places on screen too (top to bottom is the signal flow).
EQ and **COMP** buttons	Turn each section on or off. A section that is off passes the sound through.
GAIN (horizontal fader at the bottom)	Final level, –24 to +24 dB. Double-click or ⌥-click for 0 dB.

EQ (4 bands): the line shows the current frequency response (lows on the left, highs on the right), over a display of the sound's frequency content after the EQ. The four numbered circles (nodes) are the bands.

On the EQ graph	Result
Click a node	Selects that band (the row below switches to it).
Drag a node left or right	Frequency (FREQ)
Drag a node up or down	Boost or cut (GAIN); it snaps to 0 dB near 0. For Low Cut and High Cut: Q (the bump at the corner).
Scroll over a node, or ⌥-drag up and down	Q (width; higher is narrower)
Drag with ⇧	Fine adjustment.
Double-click a node	Turns that band on or off.
⌘-click a node	Resets that band's GAIN to 0 dB.

The row under the graph shows the selected band: from the left, band on/off, the band number (click for the next band), the **GAIN** (–18 to +18 dB), **FREQ** (20 Hz to 20 kHz) and **Q** (0.1 to 10) knobs, and five type buttons (Low Cut, Low Shelf, Bell, High Shelf, High Cut). For Low Cut and High Cut, **SLOPE** (12 or 24 dB/oct, how steep the cut is) replaces GAIN.

> **EQ basics** For a vocal, first set band 1 to Low Cut to remove rumble below 80 to 100 Hz. Cut a little where it sounds boxy (200 to 500 Hz), and add a little at 3 to 5 kHz for presence. Cutting usually sounds more natural than boosting.

Compressor: holds down loud passages to even out the level. The graph on the left shows input level (across) against output level (up); the vertical meter on its right shows how much the level is being reduced (gain reduction). The ◯ moving along the line is the current level; it turns orange while it is being compressed.

Knob or button	Range (default)	What it does
THR	–60 to 0 dB (–10 dB)	Threshold: sound above it is compressed.
RATIO	1:1 to 20:1 (2.0:1)	How hard: at 4:1, what goes over the threshold is reduced to a quarter.
KNEE	0 to 24 dB (6 dB)	How gently compression starts; wider is more natural.
ATK	0.1 to 200 ms (15 ms)	Attack: how quickly it reacts when the sound gets louder.
REL	5 to 2000 ms (120 ms)	Release: how quickly it recovers when the sound gets quieter.
MAKE	0 to +24 dB (0 dB)	Makeup gain: brings the compressed sound back up.
AUTO GAIN	On / off (off)	Adds makeup gain automatically from THR and RATIO.
STEREO LINK	On / off (on)	Compresses left and right equally, so the stereo image stays put.
MIX	0 to 100 % (100 %)	Blend of compressed and original sound (parallel compression).

On the compressor graph	Result
Drag left or right	THR
Drag up or down, or drag the •	RATIO (the • follows the mouse)
⌥-drag up and down	KNEE
Double-click	Resets THR, RATIO and KNEE.

> **Compressor basics** Turn COMP on and lower THR until the vertical meter moves 3 to 6 dB on the loudest parts of the vocal. Make up the lost level with MAKE (or AUTO GAIN).

MyMaximizer (maximizer)

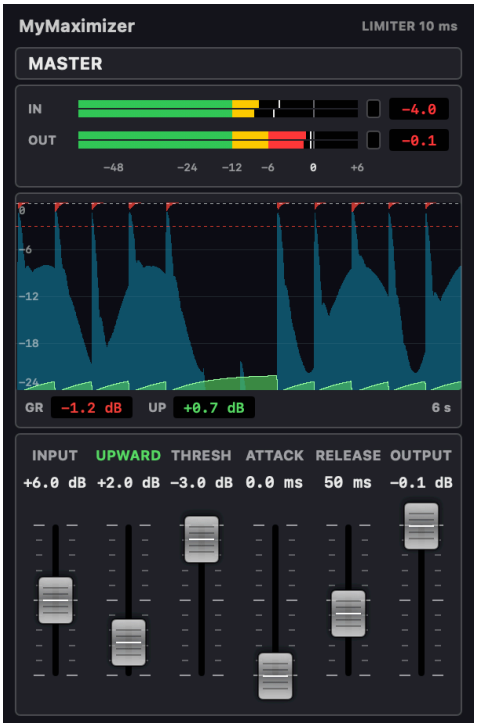


Figure 11-4 MyMaximizer

Makes the whole mix louder while keeping it under a level you set (OUTPUT). Insert it last in the MASTER post inserts.

Fader	Range (default)	What it does
INPUT	–12 to +24 dB (0 dB)	How hard the mix is driven into the limiter. Higher is louder (too high distorts and flattens the sound).
UPWARD	0 to +9 dB (+2 dB)	Upward compression: lifts quiet parts (below –6 dB) by up to this much, filling out quiet passages. 0 dB is off.
THRESH	–30 to 0 dB (–0.1 dB)	Where compression starts. Peaks above it are pressed softly (soft knee) toward OUTPUT.
ATTACK	0 to 10 ms (0 ms)	How early the level comes down before a peak. 0 ms reacts right at the peak (loudest, but distorts most easily); longer is softer.
RELEASE	10 to 500 ms (50 ms)	How quickly the level comes back. Shorter is louder but pumps and distorts more easily. UPWARD recovers 10 times more slowly.
OUTPUT	–12 to 0 dB (–0.1 dB)	Ceiling: the output never goes above it. For streaming, use –0.1 to –1 dB.

1. When inserted, THRESH and OUTPUT are both –0.1 dB, so it works as a plain limiter (prevents clipping).

2. Raise **INPUT** to set the loudness. If the red in the graph (limiting) stays above a few dB, it is too high.
3. If the peaks sound hard or crunchy, lower **THRESH**. Everything above THRESH is then pressed gradually, levelling off just below OUTPUT. About –3 to –6 dB is a good start.
4. To bring up quiet passages, raise **UPWARD**.
5. Start **ATTACK** and **RELEASE** at their defaults (0 ms / 50 ms) and lengthen them only if you hear distortion or pumping.

How THRESH works (OUTPUT –0.1 dB, THRESH –6 dB):

Peak in	–6 dB or lower	–3 dB	0 dB	+6 dB	+12 dB
Peak out	unchanged	–3.7 dB	–2.2 dB	–0.9 dB	–0.4 dB

Graph: scrolls from right to left and shows the last 6 seconds (click for 3, 6 or 12 seconds). Teal is the output level, red hanging from the top is how much the limiter is reducing the level, and green rising from the bottom is how much UPWARD is lifting it. The white dotted line marks OUTPUT, the red dotted line THRESH (when below OUTPUT). **GR** and **UP** below show the limiting and the upward lift in dB (the largest value is held for 1 second).

CLIP lamps: the squares to the right of the IN and OUT meters light red when the signal reaches 0 dB and stay lit until clicked. OUT's CLIP lamp normally never lights.

> **Processing delay** MyMaximizer looks 10 ms ahead at the peaks, so its output is 10 ms late. MyDAW compensates for this automatically, so it stays in time with everything else (chapter 10, "Automatic delay compensation"). "LIMITER 10 ms" at the top right shows this delay.

MyChorusPan (chorus, flanger, auto pan)

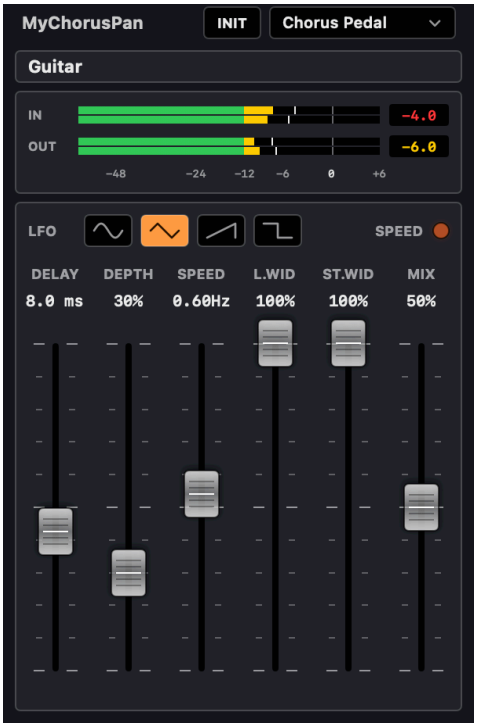


Figure 11-5 MyChorusPan

Three classic modulation effects in one. Choose the mode in the menu at the top right and press **INIT** for that mode's recommended settings. Each mode keeps its own settings, so switching modes does not lose them.

Mode	Sound	Button row	Faders
Chorus Pedal	Chorus: movement and thickness, like a guitar pedal	LFO waveform	DELAY, DEPTH, SPEED, L.WID, ST.WID, MIX
Flanger Pedal	Jet-plane sweep	LFO waveform	DELAY, DEPTH, FDBK, SPEED, L.WID, ST.WID, MIX
Auto Pan	Moves the sound from side to side	PAN waveform, SYNC row	SPEED, WIDTH

Control	What it does
INIT	Resets only the current mode to its recommended settings (the other modes stay as they are).
Waveform buttons	The shape of the movement: Sine (smooth), Triangle (even back and forth; the classic chorus), Saw (slow rise, quick return), Square (jumps between two points).
SPEED and lamp	Speed of the movement. The SPEED lamp at the top right blinks with it.
DELAY	Centre time of the moving delay: 3 to 30 ms in Chorus Pedal, 0.1 to 10 ms in Flanger Pedal.
DEPTH	How deep the movement is (0 to 100 %).
FDBK	How pronounced the flanging is (–95 to +95 %). Negative values give a hollow sound.
L.WID	Offset between the left and right movement: 0 % moves both sides together, 100 % moves them in opposite directions (widest).
ST.WID	Stereo width of the effect; 0 % makes the effect mono.
MIX	Balance of the original and the effect: 0 % original only, 50 % half each, 100 % effect only.
WIDTH (Auto Pan)	How far the sound moves; at 100 % from the far left to the far right.
SYNC, **N**, **T**, **D** (Auto Pan)	Ties the speed to the song's tempo. With SYNC on, SPEED chooses a note length (1/64 to 4 bar, 1 bar is one bar of the song's time signature) and the sound goes from side to side and back once per length. N, T and D are normal, triplet and dotted. The readout at the right shows the tempo and the speed in Hz. During playback the movement lines up with the bars (for example 1 bar with Square: right for the first half of every bar, left for the second).

Recommended settings (INIT): Chorus Pedal DELAY 8 ms, DEPTH 30 %, SPEED 0.6 Hz, MIX 50 %; Flanger Pedal DELAY 2 ms, DEPTH 80 %, FDBK +60 %, SPEED 0.2 Hz, MIX 50 %; Auto Pan Sine, 1 Hz, WIDTH 100 %, SYNC off.

1. Insert MyChorusPan on a track and choose the mode in the menu at the top right.
2. Press **INIT** to start from the recommended settings.

3. Set the strength with SPEED and DEPTH (Chorus Pedal, Flanger Pedal).

> **On an FX channel** Set MIX to 100 %: the track already plays the original, so the FX channel returns only the effect. (Insert Auto Pan on the track itself, not on an FX channel.)

> **Auto Pan level** In the centre the sound is unchanged; at either end one side is silent and the other a little (about 3 dB) louder. The combined level of both sides stays the same.

MyDeEsser (de-esser)



Figure 11-6 MyDeEsser

Softens harsh consonants in vocals, such as "s" and "sh" sounds. Only the sound above FREQ (the consonant band) is turned down, and only by the amount it goes over THRESH. Lower sounds are not changed. Insert it on the vocal track, after MyChannelStrip.

Fader / button	Range (default)	What it does
INPUT	−24 to +24 dB (0 dB)	Level going in. THRESH is compared with the level after INPUT, so you can raise it for quiet recordings.
FREQ	2,000 to 15,000 Hz (6,500 Hz)	Lower edge of the consonant band. Only sound above it is turned down. A good start is 6–8 kHz for female vocals and 4–6 kHz for male vocals.
THRESH	−60 to 0 dB (−12 dB)	The consonant band is turned down by the amount it goes over this level. The lower it is, the stronger the effect.
ATTACK	0 to 5 ms (0 ms)	How long before the level comes down. At 0 ms consonants are turned down from their very start. A little longer lets the start of each consonant through, which can sound more natural.
RELEASE	5 to 50 ms (10 ms)	How long the level takes to come back. Too long also turns down the high sounds of the next note.
STEREO LINK	On / Off (On)	On turns both sides down together, so the sound stays in place. Off turns each side down separately. Lit blue when on.
LISTEN	On / Off (Off)	Plays only the consonant band. Use it to set FREQ, then turn it off. Lit yellow when on.

1. Turn **LISTEN** on and move **FREQ** until the "s" sounds are clearest and the vowels are barely heard. Then turn LISTEN off.
2. While the vocal plays, lower **THRESH** until **GR** moves by about 3–8 dB on the "s" sounds only. If GR also moves on vowels, THRESH is too low.
3. If the vocal starts to sound lispy, raise THRESH or lengthen **ATTACK** a little.

Graph: pitch from left to right, level change from top to bottom. The red dotted line is FREQ, and the area to its right is the consonant band. The white line shows how the sound is changed right now: while consonants are being turned down, the right side dips and the dip is filled red. Drag the graph sideways to move FREQ.

SIB: level of the consonant band (left on top, right below). The red vertical line is THRESH, and the part above it turns red. Drag the red line sideways to move THRESH. **GR**: how much the consonants are being turned down (0 dB at the right end, −24 dB at the left). The numbers on the right are the largest values since you last clicked them.

The IN meter shows the level after INPUT.

> **Processing delay** MyDeEsser looks 2 ms ahead, so it starts turning a consonant down before it arrives. Its output is therefore 2 ms late, but MyDAW compensates for this automatically, so it stays in time with everything else (chapter 10, "Automatic delay compensation").

MyTuner (tuner)



Figure 11-7 MyTuner (A2, 6 cents flat)

A tuner for your guitar or other instruments. It is not inserted on a track: open it with the transport bar's **Tuner** (tuning fork) button (figure 3-2, item 10), and press the button again or the window's close button to close it. It does not change the sound, so you can use it while playing or recording.

Item	What it does
Microphone mark and input field	The input channel the tuner listens to; click to choose. When you open the tuner and the selected track is armed (R), it listens to that track's input; otherwise to the input you chose last. With a device that has no inputs it shows "No Input".
Chromatic / Guitar	**Chromatic**: the nearest semitone (also for bass or lowered tunings). **Guitar**: the nearest of the six strings in standard tuning (E A D G B E), with the string number right of the note name.
Meter	Needle left of centre: flat; right: sharp (–50 to +50 cents). Within ±2 cents it turns green.
▶ note ◀	Note name and octave (A4 = 440 Hz; the guitar's open low E is E2). ▶ lights when flat, ◀ when sharp, and both light green when in tune.
Frequency / Difference	The pitch you play (Hz) and how far it is from the target note (cents).
STROBE	Shows moving stripes instead of the needle: they move left when flat, right when sharp, and stand still when in tune. Good for fine tuning.
CAL	Reference pitch (the frequency of A4): –/+ in 0.5 Hz steps, 430 to 450 Hz. Double-click the number to type it. Default 440 Hz.

1. Choose the input your guitar is plugged into in the input field.
2. Play one string at a time and turn the tuning peg until the needle is centred and green.

> **When the display goes out** The reading stays for a moment after the sound stops, then fades. The input level is not adjusted automatically, but any level fit for recording can be measured. The settings (input, mode, STROBE, CAL) are saved in MyDAW, not in the project.

12. MIDI and instruments

Play a MIDI keyboard or import a MIDI file to sound software instruments. MIDI records what was played (which key, when, how hard), not the sound itself, so the pitch, timing, strength and length of every note can still be changed after recording.

Creating an instrument track

1. Press **+** above the track list, or right-click a track header, and choose an instrument from **Add Instrument Track**.
2. The track is added and the instrument's window opens. The track is named after the instrument.
3. The new track is armed (R on), so you can play it from your MIDI keyboard right away.

Where	What it does
🎹 and the instrument name in the track header	Click to open the instrument's window. "Loading…" is shown while it loads.
▼ next to the instrument name	Switches to another instrument (while stopped). **No Instrument** removes it.
The 🎹 row at the top of the mixer channel	Click to open the instrument's window. Volume, pan, effects and sends work as on audio tracks.

> **Note** Audio Unit (AU) instruments can be used. The instrument's settings (its sound) are saved in the project and come back when you open it again. Instrument tracks do not record or import audio.

Separating an instrument's outputs

An instrument with several outputs (a drum instrument, for example) can give each output its own mixer channel, so that each has its own effects and level. On such instruments, the 🎹 row at the top of the mixer channel shows an outputs button.

1. While stopped, click the outputs button.
2. Choose **Separate All Outputs**, or pick the outputs you want from the list.
3. A channel for each output appears to the right of the instrument's channel.

- Separated channels first go to the instrument's own channel. Its fader sets the overall level, and it goes on to MASTER. You can change the destinations (chapter 9).
- ◀ in the 🎹 row hides the separated channels and ▶ shows them again. Hiding them does not change the sound.
- To undo it all, choose **Back to the Mix** from the outputs button.

Playing a MIDI keyboard

A MIDI keyboard connected to the Mac is ready to use (also when connected later). Playing it sounds the instruments of the **armed (R on) instrument tracks**; with R off, a track no longer plays. With several armed, they all play.

Recording MIDI

1. Turn on R on the instrument track.
2. Start recording with the red **Record** button (or `R`) and play. The notes are drawn into the clip as you play.
3. When you stop, a MIDI clip remains from the bar where recording started to the bar after the last note. Nothing is kept if you played nothing.

- **With audio**: arm audio tracks too to record audio and MIDI together.
- **Overdub / replace**: switch with the 🎹 + button in the transport. With 🎹 **+ (overdub)** the notes already there stay and the new clip lies over them. With 🎹 ⇄ **(replace)**, lit red, the notes in the recorded range are removed.
- **Punch in and rollback** work as for audio (chapter 7). Notes played outside the range are not recorded.
- `⌘Z` undoes a recording.

Importing a MIDI file

Choose **File** → **Import MIDI File…**, or drag a MIDI file (.mid) onto a track's lane.

- When the current track (or the one you drop on) is an instrument track, the file goes into it. The file's other tracks become new instrument tracks; choose their instruments afterwards.
- Clips are placed where they are in the file.
- When the file's tempo or time signature differs from the project's, you are asked whether to use it (a time signature whose lower number is not 4 or 8 is ignored without asking).

Editing MIDI clips

To	Do this
Select	Click. `⇧`-click adds to or removes from the selection.
Move	Drag (on beats while snap is on). Drag up or down to move it to another instrument track.
Copy	`⌥`-drag, or `⌘C` then `⌘V` (at the playhead).
Trim	Drag the clip's left or right edge. Hidden notes do not play, and come back when the edge is moved back.
Right-click menu	Mute Clip / Unmute Clip, Duplicate Clip (at the playhead), Split at Cursor, **Merge Clips**, Delete Clip, cut, copy and paste.
Merge clips	Select two or more clips on one track, right-click and choose **Merge Clips**: they become one clip that plays the same.
Delete	`Delete`
Open in the piano roll	Double-click

The piano roll

Double-click a MIDI clip: the bottom panel switches to the piano roll and shows the whole clip. The **Mixer** / **Piano Roll** tabs on the panel's bar switch too.

Part	What it shows
Inspector (left)	Default values, note length adjustment, velocity adjustment, transpose, octave shift and swing adjustment (see "Quantize and adjustments" below).
Toolbar	Arrow, pencil and eraser tools, audition on/off (speaker), **Qtz** and the quantize modes, snap to grid, follow the track view, zoom.
Keyboard	Click a key to hear it (whether audition is on or not). C3 is MIDI note 60.
Ruler	Click or drag to move the playhead. The mouse wheel zooms in time.
Notes	The colour shows the velocity (green soft, red loud). Rest the pointer on a note to see its name and velocity. Outside the clip is shaded; notes a trim hides are drawn faint.
Lower lane	The tabs above it switch between **Velocity**, **Pitch Bend**, **Modulation** (the modulation wheel) and **Aftertouch** (how hard keys are pressed down). Velocity is the height of each line; the other three are a graph of dots where the value changes, joined by lines. The fainter line from the last dot to the clip's end shows the value carrying on. Pitch bend is 0 at the centre.
Scroll bar (very bottom)	Drag its knob (●) to move in time.

Tool / action	What it does
Arrow tool	Click selects (⇧ adds), dragging empty space selects with a frame, dragging a note moves it, ⌥-drag copies, dragging a note's left or right edge changes its length, double-clicking empty space adds a note.
Pencil tool	Click adds a note; keep dragging to set its length. It also adds on top of a note (overlaps are fine).
Eraser tool	Deletes the notes you click or drag across.
Audition (speaker button)	While on (orange), clicking a note plays it, and moving it up or down plays the new pitch.
Dragging a velocity dot	Grab the dot at the top of a note's velocity line and drag up or down to change that note alone. While you drag, the note plays every 0.5 seconds at the velocity reached, one note at a time (not when Audition is off).
Dragging in the velocity lane	Drag up or down to set the velocity of the note there; drag sideways to change the notes you pass. ⌥-drag draws a straight line and sets the notes under it along the line. With notes selected, only they change.
Editing pitch bend, modulation and aftertouch	**Pencil tool**: click or drag to draw the graph (it replaces the values over the span you draw); ⌥-drag draws a straight line. **Eraser tool**: click or drag over dots to delete them. **Arrow tool**: drag a dot up or down to change its value. Pitch bend snaps to 0 near the centre. Wheel movements on a MIDI keyboard are recorded here.
Right-click menu	Mute Notes, Unmute Notes, Cut, Copy, Paste at Playhead, Delete Notes, Select All Notes.
Follow the track view	While on, shows the same time span as the track view and scrolls with it while playing. Scrolling sideways or zooming in the piano roll turns it off.
Scrolling and zoom	Mouse wheel scrolls; ⌘+wheel zooms in time, ⌥+wheel zooms the rows.

⌘Z undoes note edits. Edits made while playing are heard at once.

The drum piano roll

With a **drum map**, tracks of that drum instrument show rows grouped by drum instead of the piano roll's keyboard. Only the order of the rows changes. The note numbers stay the same, so MIDI data made for other instruments still works. MyDAW comes with no drum maps; an instrument without one shows the normal piano roll.

A drum map is a JSON file like the one below. Put it in **MyDAW/DrumMaps** in your Documents folder and restart MyDAW.

```
{
  "subtype": "ABCD", "manufacturer": "EFGH",
  "drums": [
    { "id": "kick", "name": "Kick", "hits": [
      { "note": 36, "name": "Center" } ] },
```

```
    { "id": "snare", "name": "Snare", "hits": [
      { "note": 38, "name": "Center" },
      { "note": 40, "name": "Rim Shot" },
      { "note": 42, "name": "Choke", "isVariation": false } ] }
  ]
}
```

Item	Meaning
subtype, **manufacturer**	The instrument's Audio Unit codes (four letters or digits each).
drums	The list of drums. **id** is a unique name, **name** is what is shown, and **hits** lists its ways of striking.
note, **name**	A way of striking: its note number and the name shown.
isVariation	Optional. False makes ↑ / ↓ skip the row, which is shown in dimmer text.

On a track with a drum map, you can do the following.

Action	What it does
↑ / ↓	Moves the selected notes to another way of striking the same drum. They stop at the drum's end.
⌥ + ↑ / ↓	Does the same, and at the drum's end goes on to the next drum.
Click a drum's name	Folds that drum to one row; click again to open it. The notes do not change, and the text on each note tells its way of striking.
Click a way of striking	Plays that sound.
Drag a note up or down	Moves it to another row, also on another drum.

- Notes with numbers the kit does not use appear on **Other** rows at the bottom.
- Rows with dimmer text repeat another row's sound or are a Choke. ↑ / ↓ skip them.
- A note's number means a way of striking here, so the left panel has no Transpose or Octave Shift in this view.

Quantize and adjustments

Quantize and the adjustments change **only the selected notes**; without a selection their buttons cannot be pressed. All can be undone with ⌘Z.

Item	What it does
Qtz (`Q`)	Moves the notes to the nearest grid line. The grid is **Default Values** → **Note Length / Quantize** in the inspector; the mode buttons next to Qtz say what moves.
[|←]	Moves note starts to the grid (ends stay).
[→|]	Moves note ends to the grid (starts stay). With [|←] also on, both ends move.
[|←|]	Moves note starts to the grid, keeping their length. Turning it on turns off [|←] and [→|].
Default values	**Velocity**: of the notes the pencil adds. **Note Length / Quantize**: the length of the notes the pencil adds, also the grid for quantize and snapping (1/1 to 1/32, straight or triplet).
Note length adjustment	**Fixed Length**: sets the chosen length. **Remove Overlaps**: where notes of the same pitch overlap, the earlier one ends where the next starts (of notes of one pitch starting together, the longest is kept).
Velocity adjustment	**Fixed**: sets the value. **Scale**: multiplies by the percentage. **Offset**: adds (or subtracts) the value. Results stay within 1 to 127.
Transpose / octave shift	Moves notes in semitones. They stop where all of them can still go, so chords keep their shape. In the piano roll, `↑` / `↓` also transpose (`⇧`: an octave).
Swing adjustment	Changes each note's velocity and start by a random amount within the ranges. The start is in 1/128 of a 4/4 bar (the same length in every time signature). Lengths stay and notes stay inside the clip; overlaps are removed afterwards.

Type a value into a field and press `Return` to confirm it.

13. Tempo, time signature, click and ruler

- **Tempo (BPM)**: type a number into TEMPO (20–400, default 120). It counts quarter notes, whatever the time signature.
- **Time signature**: in METER, to the right of TEMPO, choose the upper number (1–16) and the lower number (4 or 8); the default is 4/4. A project has one time signature, with no changes during the song. Changing it does not move clips or notes: only the bar lines fall elsewhere.
- **Metronome**: turn it on to hear a click while playing or recording; the first beat of a bar is higher. With 8 as the lower number it clicks on every eighth note. Turned on during playback or recording, it starts clicking from the next beat.
- **Ruler**: button 17 in figure 3-2 switches between time and bars/beats.
- **Tempo in plug-ins**: the tempo and time signature also reach the SYNC of MyDelay and MyChorusPan and other plug-ins that can follow a tempo. Some plug-ins need their own tempo setting matched by hand.
- **Click timing and volume**: **Click timing** (ms) and **Click volume** in the settings.

Song start and end flags

You can place blue song start and end flags on the ruler (the start flag points right, the end flag left). Both are optional.

- **Place**: right-click the ruler and choose **Set Song Start Here** or **Set Song End Here**. The start must be at least one beat before the end.
- **Move**: drag the flag's pennant; the playhead does not move while you hold a flag. Clicking the pole or around it moves the playhead as anywhere else on the ruler. With snap on, the flag moves in beats. While you hold it, its position is shown next to it, in the ruler's units: bar:beat:tick (480 ticks per beat) or h:m:s.ms.
- **Overlapping the punch handles**: the punch handle moves. To move the flag instead, hold ⌥ while dragging.
- **Remove**: right-click the ruler and choose **Remove Song Start** or **Remove Song End**.

With flags in place:

- **Rewind** (←) goes to the start flag. If you are already on it or before it, it goes to 00:00.000.
- **Stop at the end flag**: playback and recording stop when they reach the end flag, and recorded clips are cut at it. Playback started after the end flag does not stop.
- **Export**: the Export Master Mix dialog starts with the flags' positions (the punch range comes first while P is on; chapter 16).

Flags are stored in beats, so they move with a tempo change.

14. Adjusting the view

Adjustment	How
Horizontal zoom	Magnifier buttons, the timeline slider (5–3200 pixels per second), the **mouse wheel over the ruler**, or a **trackpad pinch**
Height of all tracks	Track height slider (about 0.3–3×), or ⌥ + mouse wheel over the tracks. The slider scales around the current track, the wheel around the mouse pointer. Using it returns tracks resized one by one to the standard height, so all tracks scale together.
Height of one track	Drag the bottom edge of its header
Waveform height	Waveform slider (1–256×), or ⌥ ⇧ + mouse wheel over the tracks. Does not change the sound.
Mixer height	Drag the mixer's top edge

Zooming with the wheel or a pinch keeps the position under the pointer in place.

Zoomed in far, the waveform shows the individual waves (cycles) of the sound.

The timeline is as long as the song: the end of the last clip (or the song end flag) plus 5 seconds, at least 60 seconds. The part after it is shown darker; the ruler goes on to the right edge of the screen, but clicking there or playing on does not make the song longer. Recording past the end makes the song longer.

15. Saving, opening and moving projects

What a project contains

```
MySong/              ← the folder chosen in the save panel (any name)
 ├──── Ballad.mydaw     ← tracks, clip positions, mixer, plug-in settings, …
 └──── Recordings/      ← your recorded WAV files
       └──── *.wav
```

The `.mydaw` file and its folder do not need the same name. Recordings are saved in the `Recordings` folder next to the `.mydaw` file. You can rename the `.mydaw` file in the Finder.

- **Save**: ⌘S, **File → Save Project…** or the Save button. “Project saved.” appears when it is done.
- **Save As**: ⇧⌘S or **File → Save Project As…**. Type a new name and click **Save**: the project is saved as another `.mydaw` file in the **same folder**, and you go on working in that file. The original file is left as it was. The recordings are not copied, because both use the same `Recordings` folder. Saving to a different folder is not possible.
- **Open**: ⌘O or the Open button, then choose the project's `.mydaw` file. You can also click a project name in the **Recent Projects** list on the start screen, or double-click a `.mydaw` file in the Finder (MyDAW starts if it is not running). If another project is open, you are asked whether to save it first. When the project's sample rate differs from the current one, MyDAW restarts by itself and opens the project at its rate (the window stays on screen during the restart).
- **New**: ⌘N, then choose the folder and name in the save panel (see chapter 2).

> **Several projects in one folder** You can keep more than one `.mydaw` file in the same folder. They then all use the same `Recordings` folder. New recording files get the next free number, so nothing is overwritten.

Recent Projects

- Below the buttons on the start screen, up to 50 projects you have opened or saved are listed, newest first: the project name (without `.mydaw`) on the left and when it was last saved on the right.
- Hover over a project name to see where the file is. Click it to open the project.
- A project that has been moved or deleted is shown greyed out and struck through, and cannot be opened. Put it back where it was and it can be opened again.
- To take an entry off the list, right-click it and choose **Remove from List** (the project files are not deleted).

Saved with the project: track settings (name, colour, volume, pan, mute, solo, input, R/I), clip positions and edits, mixer and FX, plug-ins and their settings, sample rate, tempo, time signature, metronome, punch range, song start and end flags, and view settings.

Moving to another Mac Copy (or zip) the **whole project folder**, not just the `.mydaw` file. Install the plug-ins you use on the other Mac too. Renaming or moving the `Recordings` folder makes clips impossible to load.

Quitting Whether you choose **MyDAW → Quit MyDAW** (`⌘Q`) or close the window, you are asked whether to save (Save / Don't Save / Cancel). Cancel keeps MyDAW open.

Shrinking recordings for sharing

Before you give a project to someone else, you can rebuild the files in the Recordings folder so that they hold only what the clips play.

1. Stop playback and recording.
2. Choose **File → Optimize Recordings to Minimum Size**.
3. In the confirmation, choose what to convert with the checkboxes (all on at first): **sample rate**, **bit depth** and **channels** (stereo to mono on mono tracks). Whatever you turn off is kept as it is.
4. Click **Optimize** (or **Cancel** to stop). The project is saved automatically.
5. When it finishes, the number of files created and the total size before and after are shown. Click **OK**.

- Each clip gets its own recording file (named `Optimized_001.wav` and so on) without the parts hidden by trimming. Clips that play the same part of the same file share one file.
- Nothing is converted up. Files at or below the current project settings keep their original values.
- **Sample rate**: files at a higher rate than the project's are converted to the project's rate.
- **Bit depth**: 32-bit and floating-point files become 24-bit. Files of 24 bits or less keep their bit depth.
- **Channels**: clips on a mono track get mono files. On a stereo track, a file that was mono stays mono.
- Original files that are no longer used go to `Recordings/Unused`. Leave the Unused folder out when you share the project.
- This cannot be undone. The Undo history and copied clips are cleared.
- When other `.mydaw` files are in the same folder, nothing is done, so their recordings are not broken.

Tidying up unused recordings

Repeated punch-ins and deleted takes leave WAV files in the Recordings folder that nothing uses any more. To move them all out of the way:

1. Stop playback and recording.
2. Choose **File → Move Unused Recordings to Unused Folder**.
3. When asked that the project needs to be saved, click **Save Project and Continue** (or **Cancel** to stop).
4. A list of the moved files appears. Click **OK**.

- They go to `Recordings/Unused`. Nothing is deleted, so you can move a file back in the Finder. To free the space, delete the Unused folder in the Finder.

- Files of deleted clips are moved too. If Undo (`⌘Z`) could have brought such a clip back, the Undo history is cleared, so Undo never restores a clip whose file has gone.
- Files that a copied clip uses are kept as in use.
- When other `.mydaw` files are in the same folder, the files their projects use are kept too.
- Only WAV files directly in the Recordings folder are checked.

> **Why it saves first** The check uses the project as it is open now. Saving before moving means no file that the saved project uses is ever moved.

16. Importing WAV files and exporting

Importing a WAV file

1. Select the WAV file in Finder.
2. Drag it onto a track lane. It is added as a clip at the playhead.

The file is copied into the project's `Recordings` folder as `Import_001.wav`, `Import_002.wav` and so on (the original is untouched). Files at a different sample rate, and 16-bit or 32-bit files, are **converted automatically to a 24-bit WAV at the current sample rate** as they are copied. Converting a long file may briefly pause the screen.

Exporting the finished mix

1. Stop playback and recording.
2. Choose **File** → **Export Master Mix…**. The export dialog opens.
3. Set the file name, folder and format (see the table below).
4. Check the start and end times (seconds). They default to the punch range while P (punch in/out) is on; otherwise to the song start and end flags; without flags, to 0 and the end of the last clip.
5. Click **Export**. If a file with the same name exists, you are asked whether to replace it. Exporting takes as long as playing the song, followed by the conversion to the chosen format. The file runs exactly from the sound at the start to the sound at the end, to the sample (plug-in latency included).

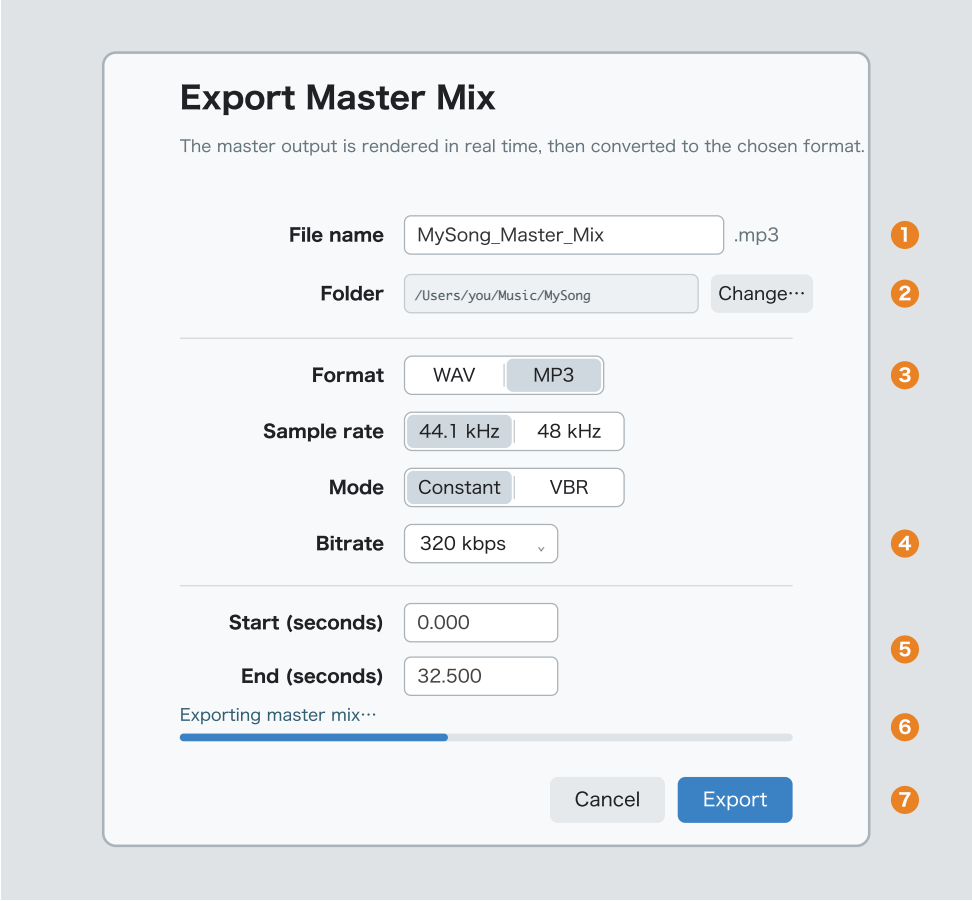


Figure 16-1 Export Master Mix dialog (MP3 selected)

1	**File name**: at first "*project name*_Master_Mix". The extension (.wav / .mp3) follows the format.
2	**Folder**: at first the project folder. Click **Change…** to choose another folder.
3	**Format** and **Sample rate**: WAV or MP3, at 44.1, 48 or 96 kHz. MP3 goes up to 48 kHz only, so 96 kHz cannot be chosen for it.
4	Settings for the format. **WAV**: bit depth 16 bit or 24 bit. **MP3**: mode "Constant" (bitrate 128 / 192 / 256 / 320 kbps) or "VBR" (quality High (V0) / Standard (V2) / Low (V4)). The default is Constant, 320 kbps.
5	**Start** and **End**: the range to export (seconds).
6	Progress: "Exporting master mix…", then "Converting…", then "Export completed successfully."
7	**Cancel** stops the export (no partial file is left). **Export** starts it.

> **Your choices are kept** The file name, folder and format are saved with the project, and the dialog opens with them next time.

17. Settings (audio and language)

Open **Settings** with the gear button (not available while playing or recording). Click **Apply** after changing anything.

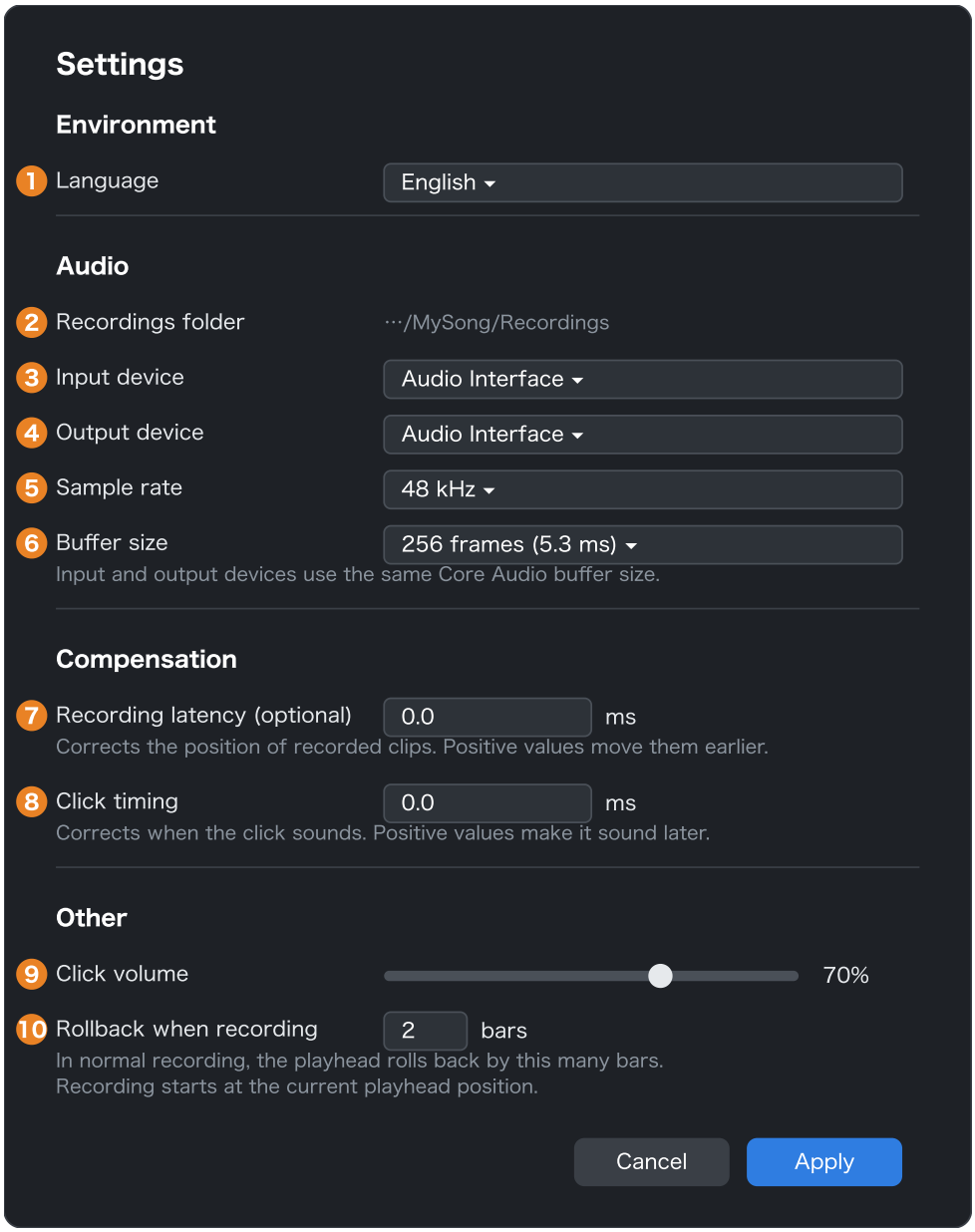


Figure 17-1 Settings

Environment	
1	**Language**: the screen language (English / 日本語). At first it follows the macOS language.
Audio	
2	**Recordings folder**: where recordings are saved (normally the project's Recordings folder).
3	**Input device**: the device you record from.
4	**Output device**: the device you listen through. It can be the same audio interface as the input or a different device (such as a monitor's speakers).
5	**Sample rate**: 44.1, 48, 88.2 or 96 kHz. Saved with each project. While the project is open, the device's own setting (the one shown in macOS Audio MIDI Setup) is set to this rate, and it is put back when MyDAW quits.
6	**Buffer size**: smaller means less delay, larger means more stability. Use 128–256 for recording and input monitoring, 512–1024 for mixing. The input and output devices use the same buffer size. The buffer size you choose is remembered and used again the next time MyDAW starts.
Compensation	
7	**Recording latency (optional)**: corrects the position of recorded clips (ms). Positive values move them earlier. Normally leave at 0.
8	**Click timing**: corrects when the click sounds (-500 to +500 ms). Positive values make it sound later.
Other	
9	**Click volume**: loudness of the click.
10	**Rollback when recording**: how many bars (1–16) the playhead rolls back when rollback recording is on (chapter 4). Recording starts at the playhead position.

Recording with a MacBook's built-in microphone and speakers The built-in microphone and speakers do not tell MyDAW how long their sound processing takes, so recordings may land about 25 ms late (with an audio interface they hardly move). Correct this with **Recording latency (optional)**, set to around **25**.

1. Set the ruler to bars and beats, turn the metronome on, and record a few bars of the click from the built-in speakers with the built-in microphone.
2. Zoom in and see how far right of the beat lines the recorded clicks start.
3. Enter that offset (ms) in **Recording latency (optional)** , click **Apply**, and record again to check that the clicks line up.

When you go back to an audio interface, set the compensation back to 0.

Changes that need a restart When you change Language, Input device, Output device or Sample rate and click Apply, MyDAW asks to restart. Choose **Save and Restart** or **Restart Without Saving** (the new settings already reach the devices when you click Apply, so you cannot go back without restarting). Even without saving, the new sample rate stays in use and is recorded in the project the next time you save. After the restart the project that was open opens again. A Buffer size change alone takes effect without a restart.

MyDAW also switches the macOS sound devices While it runs, MyDAW makes your chosen input and output the macOS default input and output, and sets the devices to the project's sample rate. Both are put back when it quits. So while MyDAW is running, other apps also play through the chosen output, at that rate. If MyDAW crashes they are not restored — set them back in System Settings → Sound and in Audio MIDI Setup.

18. Keyboard shortcuts

Key	Action
`Space`	Play / stop
`R`	Record (same as the red Record button; stops when running)
`←`	Go to the song start flag (on or before it: to the beginning)
`Delete`	Delete the selected clips (or the audio in the selected range)
`⌘X` / `⌘C` / `⌘V`	Cut / copy / paste at the playhead
`⌘A`	Select all clips
`Esc`	Clear the selection
`⌘Z`	Undo
`⇧⌘Z` / `⌘Y`	Redo
`⌘S`	Save
`⇧⌘S`	Save As
`⌘O`	Open
`⌘N`	New project
`⌘?`	MyDAW Help (opens this manual as a PDF on the web)
`⇧`-click / `⌘`-click (clip)	Add to / remove from the selection
Drag (empty spot) / `⇧`-drag	Marquee select / add with a marquee
`⌘`-drag (tracks)	Select a time range
`⌥`-drag (clip)	Duplicate while moving
Double-click (fade diamond)	Reset the fade shape to Auto
Mouse wheel over the ruler / pinch	Horizontal zoom
`⌥` + mouse wheel	Track height
`⌥` `⇧` + mouse wheel	Waveform height
`⌥`-click (mixer)	Fader / send to 0 dB, pan to centre
`⌘`-drag (mixer)	Fine adjustment
Double-click (values, names)	Type a value / rename
Double-click (MIDI clip)	Open it in the piano roll

After you click the piano roll, these keys work there (click elsewhere to go back to the tracks).

Key	Action (piano roll)
`Delete`	Delete the selected notes
`↑` / `↓`	Move the selected notes up / down a semitone (`⇧`: an octave). In the drum piano roll: to another way of striking the same drum (`⌥`: also on to the next drum)
`Q`	Quantize the selected notes
`⌘X` / `⌘C` / `⌘V`	Cut / copy / paste notes at the playhead
`⌘A`	Select all notes inside the clip
`⌥`+drag (note) / (velocity lane) / (pitch bend and other lanes, pencil)	Copy and move / set velocities along a straight line / draw a straight line
`⌘`+wheel / `⌥`+wheel	Zoom in time / zoom the rows

`Space`, `R` and `←` also work while you are using a plug-in window (except while typing a value).

19. Troubleshooting

Problem	What to check
Cannot record / meter does not move	Microphone permission (System Settings → Privacy & Security → Microphone), Input device, the track's R button, the input channel.
No sound	Output device, MASTER and track faders, M (mute) / S (solo), speaker or headphone volume.
Play does not record	Recording uses the red **Record** button; Play only plays.
Recordings are out of time	Check the buffer size, record the same sound again to measure the offset, and correct it with Recording latency (optional).
The sound breaks up / ● Dropout appears in the status bar	Raise the buffer size in Settings (512–1024). Remove heavy plug-ins (turning them off does not lower the load). The status bar's CPU meter shows whether the load went down.
Effected input sounds late	Buffer size 128–256; turn down the interface's direct monitoring.
A plug-in is missing from the list	Effects are listed for inserts; instruments under **Add Instrument Track**. AU effects driven by MIDI and VST3 instruments are not listed. Products with an AU version are listed as AU only.
Playing the MIDI keyboard makes no sound	Check that the instrument track's R is on, that an instrument is chosen (the 🎹 name in the track header) and that the keyboard is connected to the Mac.
Cannot add a plug-in	Stop playback first.
A plug-in name is red and its window does not open	Hold the pointer over the name to see why. If it "cannot be used", replace it with another plug-in such as the stereo version; if it "was not found", install the plug-in and open the project again.
Cannot import a WAV	Check that the file is an intact audio file such as WAV. Other sample rates and bit depths are converted automatically.
No sound from the chosen output device	After changing devices, restart MyDAW from the dialog. The new device is used only after the restart.
Other apps also play through MyDAW's output	While MyDAW runs, the chosen devices are the macOS defaults. They are put back when MyDAW quits.
After a crash, the Mac still uses MyDAW's devices	Set output and input back in System Settings → Sound.
Playback and recording suddenly stop, and the input stays silent afterwards	This happens when the input or output is changed in System Settings → Sound while MyDAW is running. Restart MyDAW.

Another app's recording (such as a screen recording) is full of loud noise	Launch MyDAW before you start recording, and quit it only after recording ends. MyDAW switches the macOS input and output when it launches and quits.
A clip says its file cannot be found	Right-click → Choose Recording File… and pick the WAV again.
You want the screens in Japanese (or English)	Choose it under Language in Settings, click Apply and restart.

20. Glossary

Term	Meaning
DAW	Digital Audio Workstation — software for recording, editing and mixing.
Track	A lane that records and plays one instrument or part.
Clip	One recorded piece of audio that you can move and edit.
Record arm	Making a track ready to record with its R button.
Punch in/out	Re-recording only a chosen range.
Crossfade	Joining two sounds by fading one out while fading the other in.
Fade curve	The shape in which the level changes during a fade: linear, equal power, bowed up, sagging, and so on.
Equal power	A fade shape that keeps the perceived level steady through a crossfade.
Range selection	Selecting a stretch of time rather than whole clips.
Normalize	Raising or lowering a level so the loudest sample reaches full scale (0 dBFS).
Fader	The control that sets volume.
Pan	Where the sound appears between left and right.
dB (decibel)	Unit of level. 0 dB is the reference; -6 dB is roughly half as loud.
Mute / solo	Silence a track / hear only that track.
Folder	Holds several tracks so they can be moved, muted, soloed and folded together.
Current track	The selected track. New tracks and pastes are placed from it.
Plug-in (effect)	Add-on software that processes sound; formats AU and VST3.
INSERT	Effects placed directly on a channel; sound passes through them top to bottom.
SEND / FX channel	Sending part of a track's sound to a shared effect (the FX channel).
MASTER	The final output where all tracks are combined.
Buffer size	The block size audio is processed in. Smaller means less delay but more CPU load.
Sample rate	How many times per second sound is measured (44.1 / 48 / 88.2 / 96 kHz).
Input monitoring	Listening to the incoming sound through the track's effects.
MIDI	A record of what was played on a keyboard (pitch, timing, strength, length), not the sound itself, so it can be changed freely afterwards.
Instrument	A plug-in that plays sound from MIDI: piano, drums, bass and so on.
MIDI clip / note	A piece of MIDI performance / each single note in it.

Velocity	How hard a key was played (1 to 127).
Pitch bend / modulation / aftertouch	The wheel that bends the pitch while playing / the wheel that adds vibrato and similar effects / how hard a held key is pressed further down. MIDI information separate from the notes.
Piano roll	The editor with keys running up the side and time across, where notes are edited.
Quantize	Moving notes onto the grid (note divisions).